# SAMUEL KWAME YEBOAH

Quantitative Finance · Risk Analytics · Fintech

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**Portfolio:** samuelyeboah432-ai.github.io/portfolio · **GitHub:** github.com/samuelyeboah432-ai
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## PROFILE

Finance graduate (MPhil, University of Ghana Business School; BA Finance and Mathematics) who combines quantitative risk modelling with hands-on fintech product work. I have built working prototypes for mobile money fraud detection and credit scoring of informal traders, and I bring several years of credit operations experience in Ghanaian banking and microfinance. I work in R, Python and Excel, and I explain technical results in plain language for decision makers.

## FINTECH PROJECTS

**Check Am · Mobile money scam detection** 2026
Live demo: samuelyeboah432-ai.github.io/check-am · Code: github.com/samuelyeboah432-ai/check-am

- Designed a Ghana-first scam checker that scores suspicious SMS, WhatsApp and investment messages and names the scam type (MoMo reversal, fake promo, Ponzi, fake job, phishing) with plain-language red flags and next steps
- Built a weighted pattern library of local scam scripts with a rule that no verdict is ever shown as fully safe
- Wrote the business case: a free consumer tool that grows a community scam database, monetised as a fraud-screening API for fintechs, MFIs and telcos

**Adwuma · Bookkeeping and loan-readiness scoring for market traders** 2026
Live demo: samuelyeboah432-ai.github.io/adwuma · Code: github.com/samuelyeboah432-ai/adwuma

- Prototyped chat-based bookkeeping that turns plain-language entries into sales and expense records and a live cash position
- Designed a transparent loan-readiness score from record consistency, net cash margin, sales stability and stock spending, with an indicative monthly repayment capacity for loan officers
- Mapped the lender referral model against the Bank of Ghana digital credit directive, the Credit Reporting Act, 2007 and the Data Protection Act, 2012

**Other builds** 2025 – 2026

- **APEX:** AI-assisted investment research tool producing sentiment, trade signals, fundamentals and risk notes across forex, equities and commodities
- **Gold Zone:** rules-based XAUUSD trading strategy with a documented backtest and a forward-test log
- **Portfolio risk workbooks:** formula-driven Excel models of returns, beta, parametric and historical VaR and CVaR, and Jarque-Bera tests on Ghana Stock Exchange data

## PROFESSIONAL EXPERIENCE

**Research Assistant** 2022 – present
*University of Ghana Business School, to Dr Emmanuel Joel Aikins Abakah*

- Build and validate daily datasets across 13 to 15 markets from Bloomberg, Datastream and public sources, with cross-exchange date alignment and gap resolution
- Estimate TVP-VAR, CAViaR, wavelet correlation and quantile regression models in R and Python, and produce tables, figures and commentary for four concurrent manuscripts with co-authors in Ghana, South Africa, China and the US

**Graduate and Teaching Assistant, Finance** Sept 2023 – Sept 2025
*University of Ghana Business School*

- Tutorials and assessment for Financial Econometrics, Financial Engineering with Derivatives, Financial Management, Financial Risk Modelling and Digital Finance
- Designed a risk metrics teaching module with Excel models on rolling volatility, CVaR, scenario analysis and stress testing, and set examination questions and marking schemes

**Credit Operations** 2014 – 2021

*Zenith Bank Ghana (2020 – 2021) · Letshego Ghana (2018 – 2019) · SIC Life Savings and Loans (2014 – 2016)*

- Prepared loan documentation, reviewed credit files and monitored accounts across a commercial bank, a microfinance lender and a savings and loans company
- Gained first-hand view of how lenders assess informal and salaried borrowers, which shapes the credit scoring design in Adwuma

## RESEARCH

**Is it the export or the dollar? Commodity tail risk in African stock and currency markets.** Yeboah, S. K. *Under review.*

- Eight African markets, 2015 to 2026. Own-commodity crashes coincide with equity crashes more often than placebo commodity crashes after VIX and dollar controls, while currencies follow a common dollar factor
- Shows that standard quantile VAR connectedness overstates tail spillovers and gives a centred-covariance correction

**MPhil thesis:** Tail risk connectedness between African stocks, global commodities and Islamic assets (2014 to 2024). Connectedness is U-shaped across quantiles, so diversification weakens in market extremes. Extended to optimal hedge ratios and portfolio weights.

**Contributions:** data and estimation for four real estate finance papers with Akinsomi and Abakah, including AfRES 2025 Conference Proceedings.

## EDUCATION

**MPhil in Finance** 2022 – 2026

*University of Ghana Business School, Legon · CGPA 3.41/4.0*

**BA in Finance and Mathematics** 2017 – 2021

*University of Ghana, Legon*

## SKILLS

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| **Risk and credit** | Credit analysis and scoring, VaR and CVaR, stress testing, scenario analysis, hedge ratios, fraud risk |
| **Quantitative** | Time-series econometrics, GARCH and DCC-GARCH, quantile methods, connectedness, cointegration, bootstrap inference |
| **Tools** | R, Python (pandas, NumPy, statsmodels), Excel modelling, EViews, HTML and JavaScript prototyping, Git and GitHub, LaTeX |
| **Data** | Bloomberg, Datastream, FRED, World Bank, central bank and exchange sources |
| **Product** | Problem framing, prototyping, business models, fintech regulation in Ghana |

## REFEREES

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